Our first Unikernels in OCaml — *for fun and profit*!

Lambda Nantes - Workshop 4

April 2026

Lambda Nantes

- **One meetup per month** (Thanks Arnaud)
- We are **always** looking for speakers (for presentations or workshops)
- Despite what the name (*Lambda*) might suggest, **we're open to all kinds of topics** (with a particular interest in Applicative languages, software engineering and formal method).

Feel free to **join us** as a speaker (and obviously as a participant).

The Workshop

Hello! In this **interactive presentation** (*a workshop*), we'll get started with creating **unikernels** using the wonderful OCaml language.

In my opinion, this is a topic that's very easy to summarize:

The Workshop

Hello! In this **interactive presentation** (*a workshop*), we'll get started with creating **unikernels** using the wonderful OCaml language.

In my opinion, this is a topic that's very easy to summarize:

> *A unikernel is **a single-application VM** image where the application and **minimal operating system** are compiled into one optimized binary.*

The Workshop

Hello! In this **interactive presentation** (*a workshop*), we'll get started with creating **unikernels** using the wonderful OCaml language.

In my opinion, this is a topic that's very easy to summarize:

> *A unikernel is **a single-application VM** image where the application and **minimal operating system** are compiled into one optimized binary.*

But **very hard** summarize in a workshop (I was being very ambitious!)

Plan

- What are *Unikernels*
  - What are they good for?
  - Why are they not as good?
  - Unikernels in OCaml
    - Mirage and Mkernel
    - Solo5
    - Actors in the world of Unikernels in OCaml

Plan

- What are *Unikernels*
  - What are they good for?
  - Why are they not as good?
  - Unikernels in OCaml
    - Mirage and Mkernel
    - Solo5
    - Actors in the world of Unikernels in OCaml
- Our first Unikernel (*Quick Demonstration*)
  - Hello World (Unikernel that just print “Hello World”)
  - Hello World as a server (Unikernel that always respond “Hello World”)
  - A first webserver (using a Webframework)
  - A Webserver with HTML (crunching files)
  - Free Project (The fun part — and it might take a while)

Plan

- What are *Unikernels*
  - What are they good for?
  - Why are they not as good?
  - Unikernels in OCaml
    - Mirage and Mkernel
    - Solo5
    - Actors in the world of Unikernels in OCaml
- Our first Unikernel (*Quick Demonstration*)
  - Hello World (Unikernel that just print "Hello World")
  - Hello World as a server (Unikernel that always respond "Hello World")
  - A first webserver (using a Webframework)
  - A Webserver with HTML (crunching files)
  - Free Project (The fun part — and it might take a while)

The first four unikernels will mainly be presentations (which you can follow interactively), and the last one will be the actual open-source project—and ***OCaml experts*** *can lend a hand!*

Why

Because I love promoting OCaml and...

Why

Because I love promoting OCaml and...

I'd really love for us to finally create a website for LambdaNantes, and I like to imagine it as **a galaxy of Unikernels surrounding a static site**, allowing us to develop libraries that are useful to the community (like `ocaml-biscuit`).

Why

Because I love promoting OCaml and...

I'd really love for us to finally create a website for LambdaNantes, and I like to imagine it as **a galaxy of Unikernels surrounding a static site**, allowing us to develop libraries that are useful to the community (like `ocaml-biscuit`).

- Easy to host/deploy
- Funny to maintain
- A beautful **over-engineered** approach!
- **We'll probably need to get organized** — if that sounds good to you — in a smart way :)
- A nice community project

So, what are Unikernels ?

So, what are Unikernels ?

*A unikernel is a **specialized**, **single-purpose machine image** that links application code with **only** the operating system components it actually needs.*

So, what are Unikernels ?

*A unikernel is a **specialized**, **single-purpose machine image** that links application code with **only** the operating system components it actually needs.*

*The result is a **single bootable image** that runs directly on a hypervisor, microVM, or bare metal target. There is no shell, no unnecessary drivers, and no multi-user support — only the application and the minimal runtime required to execute it.*

So, what are Unikernels ?

*A unikernel is a **specialized**, **single-purpose machine image** that links application code with **only** the operating system components it actually needs.*

*The result is a **single bootable image** that runs directly on a hypervisor, microVM, or bare metal target. There is no shell, no unnecessary drivers, and no multi-user support — only the application and the minimal runtime required to execute it.*

So yes, we're going to build an operating system for a single application

The main difference with containers

With a Unikernel, our application doesn't need a full operating system to run; in terms of execution, it might superficially resemble containerization?

The main difference with containers

With a Unikernel, our application doesn't need a full operating system to run; in terms of execution, it might superficially resemble containerization?

Container	Unikernel
Shares the host Linux kernel	Has its **own** minimal kernel image
Lightweight process isolation	Stronger isolation
	Even smaller runtime surface
Great tooling ecosystem	
isolated process	isolated single-purpose mini-OS